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Guide to DIN EN 12464-1

Lighting of work places –
Part 1: Indoor work places

Contents

2nd corrected edition

Foreword

This Guide is designed to facilitate the application of the newly revised DIN EN 12464-1 “Lighting of work places – Indoor work places” (August 2011) for the planning and design of lighting installations. In Germany, DIN EN 12464-1 often needs to be applied alongside workplace regulation ASR A3.4 “Beleuchtung” (Lighting). In certain instances, the two differ in nomenclature and content.

This Guide sets out to show how planners and designers can meet the requirements of both DIN EN 12464-1 and ASR A3.4.

European standard EN 12464-1 is a product of detailed discussion. Like the preceding edition published in March 2003, it covers all the relevant indoor applications. However, it has been revised and extended in a number of places. Published in August 2011, it documents the state of the art. EN 12464-1 applies throughout Europe and – like ISO 8995/CIE S 008 – as an ISO standard worldwide. It has been published in Germany as national standard DIN EN 12464-1 with a national foreword.

The terms used in the standard are explained here in plain English and set against the corresponding terms used in ASR A3.4. Lighting designs can be created on the basis of DIN EN 12464-1 but because of varying assumptions they are not necessarily comparable. This Guide helps permit comparability by recommending maintenance factors, for example, and by showing how reference surfaces can be defined. The recommendations and examples are selected so that designs can meet the requirements of both DIN EN 12464-1 and ASR A3.4. They are also broadly compliant with the statutory occupational accident insurers’ office lighting guide BGI 856 “Beleuchtung im Büro” (Version 2.0 2008-10), which in turn is based on the March 2003 edition of DIN EN 12464-1 and core elements of DIN 5035 Part 7 “Lighting of interiors with visual display work stations” (August 2004).

This Guide explains the terminology and application of DIN EN 12464-1 and ASR A3.4 but it is no substitute for careful study of the two sets of rules.

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and

LiTG

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1. What is new in DIN EN 12464-1

The revised version is basically structured along the same lines as the original DIN EN 12464-1 published in March 2003. The new terms introduced in that edition – terms not contained in the old DIN 5035 Parts 1 and 2 – were explained in the ZVEI Guide published in April 2005.

The new DIN EN 12464-1 places a clearer emphasis on the importance of daylight and the requirements it contains generally apply to both daylight and artificial lighting. Where requirements apply to only one or the other, the fact is specifically pointed out:
- glare rating by the UGR method applies only to artificial lighting
- uniformity specifications do not apply to daylight from the side

The revised standard also contains additional criteria and methods:
- Differentiation of the maintained illuminance uniformity (U_o) required for the task area, activity area or interior area in an additional column in the tables presented in section 5.3
- Definition of a “background area” in addition to the task area and the immediate surrounding area
- Introduction of cylindrical illuminance and modelling as criteria for assessing lighting in the interior space
- Wall and ceiling illuminance requirements for balanced luminance distribution
- Definition of an illuminance grid in line with DIN EN 12464-2
- Update of luminance limits permissible for luminaires to take account of current display screen technology

DIN EN 12464-1 lists the lighting criteria that remain vital for lighting quality:
- Agreeable luminous environment
- Harmonious luminance distribution
- Adequate illuminance for the interior areas, task areas or activity areas listed in the tables “Schedule of lighting requirements”
- Good uniformity
- Limitation of direct and reflected glare, including veiling reflections
- Correct directionality of lighting and agreeable modelling
- Appropriate colour rendering and colour appearance of the light
- Avoidance of flicker and stroboscopic effects
- Quality of daylight
- Variability of light

DIN EN 12464-1 repeatedly points out that lighting should be designed to permit control or regulation. This means that an effective lighting management system should be used.

The criteria “colour rendering” and “colour appearance” are not covered in more detail. Basically, the new standard regards $R_a > 80$ as a minimum requirement for constantly manned work stations and $R_a > 90$ for work stations with special colour matching requirements.

Lighting parameter symbols

DIN EN 12464-1 contains a number of lighting parameter symbols that are in general use:

$\bar{E}_m$ = (average) maintained illuminance
$\bar{E}_z$ = mean cylindrical illuminance
$\bar{E}_v$ = average vertical illuminance
UGR_L = UGR limits for rating glare
U_o = uniformity, corresponds to g1
R_a = colour rendering index

[01] Correct desk lighting – user-friendly, tailored to requirements and coordinated with daylight – makes for an agreeable workplace.

2. Statutory situation in Germany

DIN EN 12464-1 in relation to the Ordinance on Workplaces (Arbeitsstättenverordnung), workplace regulation ASR A3.4 and retracted regulatory instruments

Basic lighting requirements relating to the health and safety of people at work are regulated in Germany by the workplace ordinance "Arbeitsstättenverordnung" (ArbStättV). All work premises fall within the scope of this ordinance. The general lighting requirements of the ArbStättV are further concretised in the workplace regulation ASR A3.4 "Beleuchtung" (Lighting).

Other sector-specific references to lighting are found in statutory accident insurers' publications. The accident prevention regulation "Grundsätze der Prävention" (BGV A1 or GUV V A1) refers to the ArbStättV and applies additionally to persons who are voluntarily insured.

In consultation with clients, lighting designers need to observe good engineering practice standards, which in Germany are set out in DIN EN 12464-1.

The following regulations referred to in the April 2005 guide are no longer applicable or referenced: ASR 7/3, DIN 5035 Parts 1 and 2, BGR 131.

2.1 Additional and differing requirements

If lighting installations in work premises are designed and/or operated only in compliance with DIN EN 12464-1, they may not meet the aforesaid statutory minimum requirements in Germany or the lighting requirements set out by the statutory accident insurance institutes. Additional or differing requirements need to be met, in particular, with regard to:

- the way task areas are combined to form a work station
- the extension of the immediate surrounding area to include the rest of the room
- the level of horizontal illuminance for certain work stations
- minimum vertical and cylindrical illuminance
- uniformity of illuminance

To meet the goals of occupational health and safety, deviations from ASR A3.4 need to be assessed for risk.

ASR A3.4 requires a daylight quotient of at least 2%, a minimum of 4% where skylights are used or a ratio of glazed area (windows, doors, walls, skylights) to floor area of at least 1:10 (approx. 1:8 shell dimensions). Work stations should preferably be positioned near windows.

Designs based on this Guide conform to DIN 12464-1 and ASR A3.4

Terms and methods are interpreted in this Guide to DIN EN 12464-1 so that the intentions of ASR A3.4 are also taken into account. Work stations designed in line with the recommendations of this Guide thus meet the requirements of both DIN EN 12464 1 and ASR A3.4.

2.2 Maintained illuminance $\bar{E}_m$

Illuminance levels impact significantly on the speed, ease and reliability with which visual tasks can be performed. The illuminance values specified in the standard are maintained values, i.e. values below which the average illuminance on a reference surface should not fall. In other words, they are the average illuminance values reached when maintenance needs to be carried out.

The tables in section 5.3 of DIN EN 12464-1 show the maintained illuminance values required for task areas, activity areas and interior areas. Appendix 1 of ASR A3.4 lists minimum values for work rooms, work stations and activities *(cf. Appendix 2: “Differences between DIN EN 12464-1 and ASR A3.4”, page 34 f.)*.

Maintained illuminance = minimum illuminance

“Maintained illuminance” is defined in DIN EN 12464-1 as the level of illuminance below which the average illuminance on a reference surface must not fall.
It is thus identical to the “minimum illuminance” defined in ASR A3.4.

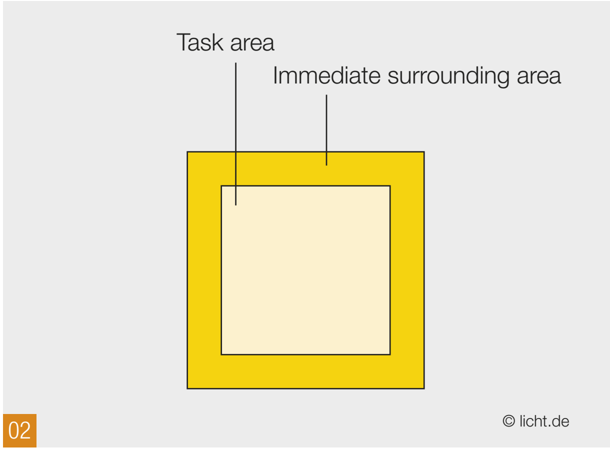
3. Work stations

Task area, immediate surrounding area and background area

DIN EN 12464-1 requires the right task lighting in the right place.

The **task area** is defined as the area in which the visual task is carried out. The visual performance required for the visual task is determined by the visually relevant elements (size of objects, background contrast, luminance of objects and presentation time) of the activity performed. The task reference surface can be horizontal, vertical or inclined.

The **immediate surrounding area** is defined as a band surrounding the task area within the field of vision. It needs to be at least 0.5 m wide.



[02] Task area and immediate surrounding area according to DIN EN 12464-1

Task area corresponds to work station area

In ASR A3.4, the reference surface analogous to the task area is known as the work station area. The work station is made up of work space, movement space and all ancillary space used for work-related tasks (see Fig. 5).
For the sake of simplicity, this Guide generally refers only to the “work station area”.
Another ASR requirement is that the adjoining surrounding area should extend to the walls of the room or to adjacent circulation routes.

Symbols in DIN EN 12464-1 and ASR A3.4

Both in DIN EN 12464-1 and in ASR A3.4, uniformity is defined as the ratio of the lowest to the average illuminance value in the illuminance grid. DIN EN 12464-1 – in line with other European and international standards – uses the symbol U_0.

Why is uniformity shown to the second decimal place in DIN EN 12464-1?

When limits are quantified, the figures are normally rounded. This means that a value of 0.5 stands for all values between 0.45 and 0.54. DIN EN 12464-1 adds an extra decimal place for greater accuracy: 0.50 stands for the narrower range of 0.495 to 0.504.

Uniformity requirements of ASR A3.4

ASR A3.4 requires 0.6 uniformity for the work station area and stipulates that the lowest illuminance should not be in the area where the primary visual task is performed. The uniformity required in the surrounding area is 0.5. This means that uniformity requirements are always higher for the surrounding area and sometimes higher for the work station area than for the equivalent areas in DIN EN 12464-1 (immediate surrounding area and task area).
Work station lighting should be designed to meet the uniformity requirements of ASR A3.4.

Defining the task area and the immediate surrounding area gives the designer the freedom to create a lighting design based on the visual requirements for a particular activity within a given space. It needs to be remembered that some visual tasks may extend over large areas.

The designer is thus required to document the size and location of the task area(s).

If the size and/or location of the task area are not known, DIN EN 12464-1 stipulates that either the whole room (or room zone) should be assumed to be the task area or the whole room should be uniformly illuminated at a level defined by the designer. When the task area is known, the lighting installation needs to be modified to achieve the relevant illuminance levels required.

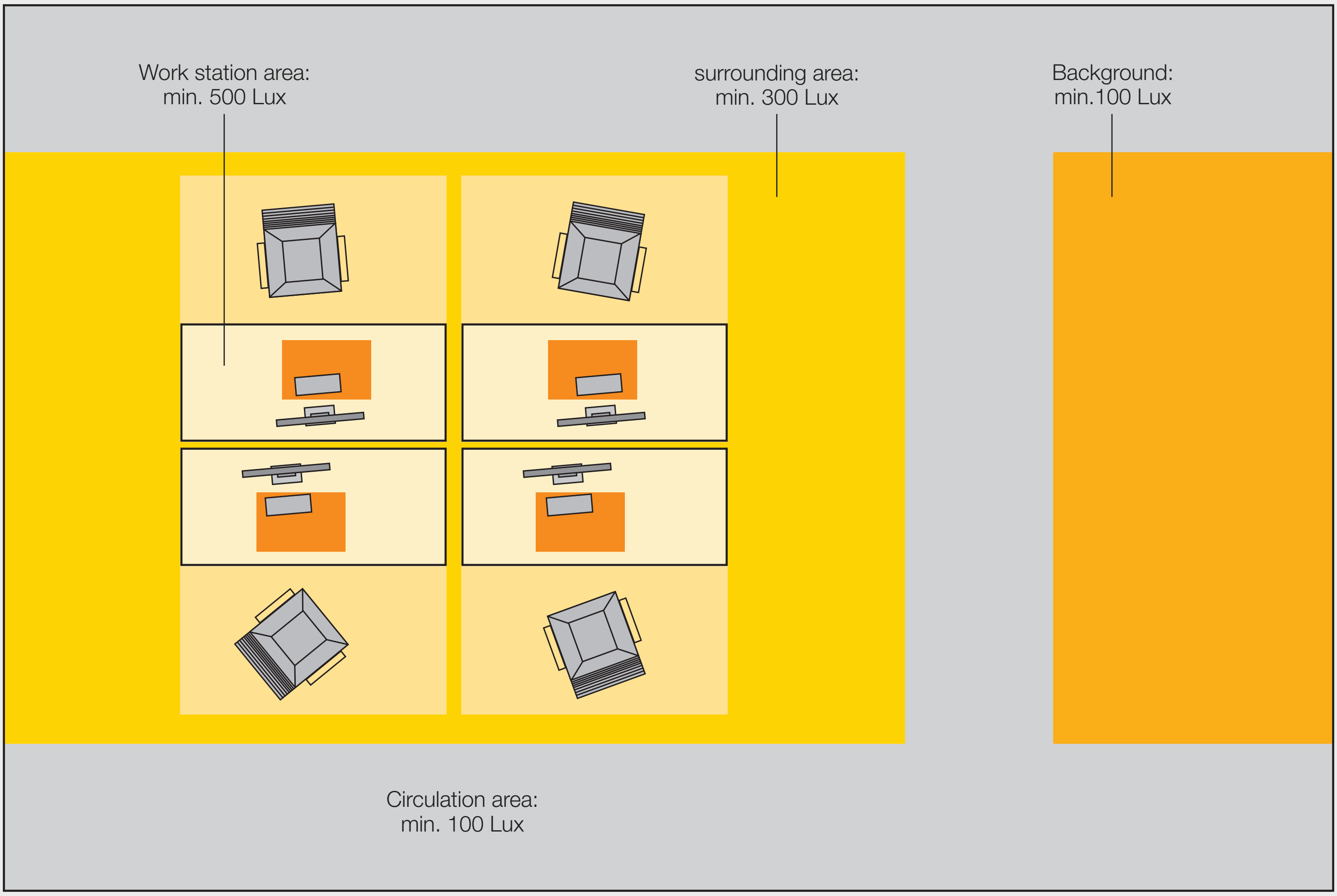
ASR A3.4 is more specific here, defining the work station area as an area in which visual tasks may be presented. For illuminances up to 500 lux, maintained illuminance needs to be observed across the work station area; for illuminances over 750 lux, it should be observed on the work surface.

The surrounding area borders directly on one or more work station areas and from there extends to the walls of the room or to circulation routes.

In very large rooms where work stations are occasionally or regularly not manned (e.g. in a call centre), DIN EN 12464-1 allows a **background area** to be applied *(see Fig. 03)*. It should be seen as a strip at least 3.0 m wide.

The maintained illuminance required for surrounding and – where applicable – background areas depends on the requirements that need to be met in the work station area.

Illuminance uniformity
The tables in section 5.3 of DIN EN 12464-1 show the uniformity (U_o) required for task areas, activity areas and interior areas. For immediate surrounding areas and background areas, the stipulated uniformity U_o is 0.40 and 0.10 respectively.



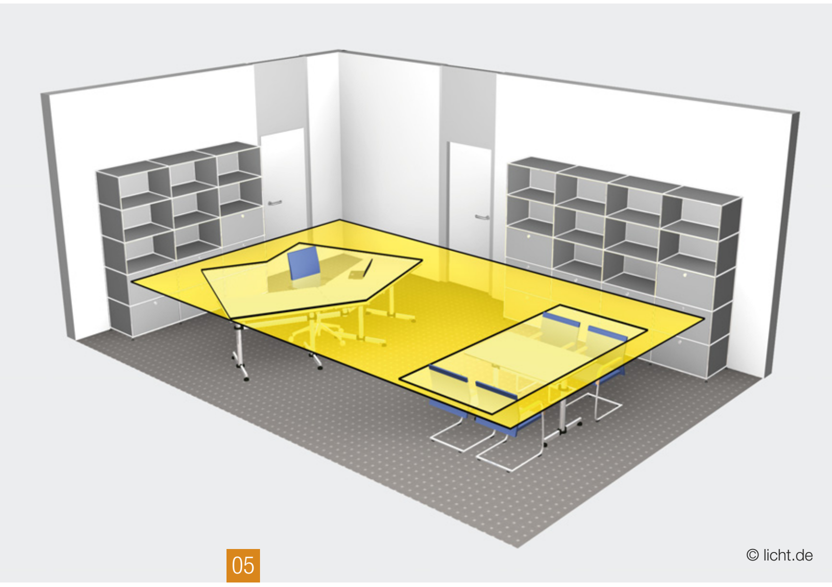
03

[03] Typical plan of work station area, surrounding area, circulation zone and adjoining background area in a very large room (e.g. call centre, industrial building)

3.1 Definition of work station areas

04

[04] The work station area consists of working space (light yellow) and user space (medium yellow) as well as the ancillary space used for tasks directly related to the work (ASR A3.4). Typical dimensions: 1.8 m x 1.8 m



05

[05] Office work station area: “display screen work” (medium yellow, left), “meeting” (medium yellow, right) and “surrounding area” (dark yellow); reference height for illuminance: 0.75 m above floor level

- Areas where different visual tasks may be performed normally form a group of interconnected surfaces comprising work space, movement space and ancillary space used for tasks directly related to the activity. Visual tasks may also be vertical or inclined. They can be grouped to form an area of the work station, which generally encompasses a horizontal surface *(see also Fig. 03 and Fig. 04)*.
- Task areas on vertical or inclined surfaces should be considered a work station area if the visual tasks performed there require more than just brief attention. Illuminance needs to be determined according to the angle of inclination. In the case of a whiteboard, for example, vertical illuminance should be used.
- Illuminance calculations for work station areas and surrounding areas can ignore a marginal strip extending 0.5 m from the walls. It needs to be ensured that no part of the work station area projects into the strip. If that is the case, the marginal strip may not always be ignored at the point(s) in question *(see also Fig. 16, page 18)*.

ASR A3.4 divides **lighting concepts** into

- **room-related lighting**, where the arrangement of work stations is unknown or flexible;
- **task area lighting**, where the arrangement of work stations is known or the nature of work stations diverse;
- **work surface lighting**, where special visual tasks are performed or lighting is individually adapted to meet the visual requirements of employees.

The application of these concepts is in accordance with the design objectives of DIN EN 12464-1.

How big is a work station area in an office?

The minimum dimensions of an office desk are 1.6 m x 0.8 m. Added to this are movement space and ancillary space (DIN 4543-1). In many cases, the actual size of furniture is unknown at the time of planning. It is recommended that the work station area should be assumed to be 1.8 m x 1.8 m square *(see also Fig. 04)*.

3.2 Examples of how work station areas can be taken into account by the lighting designer

a. Offices

Offices can accommodate one or more work stations in known or unknown arrangements. A work station area includes desktop surface(s) and user space. The working plane is assumed to be 0.75 m above floor level.

a.1 Office with single work station

The position of the workstation is known. The surrounding area is taken to be the rest of the room less a 0.5 m wide marginal strip.

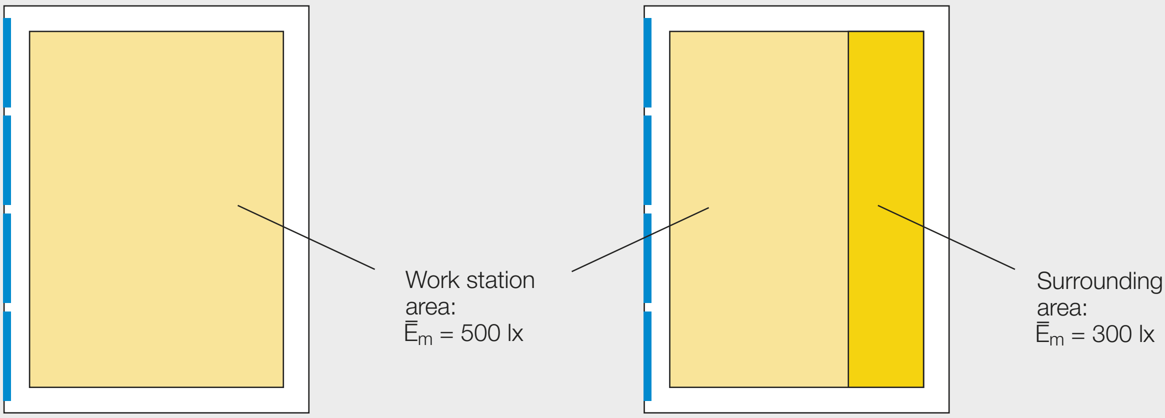
a.2 Office with unknown arrangement of work stations

If the arrangement of work stations is completely unknown, the work station area should be taken as the whole room less a 0.5 m wide marginal strip, which is ignored.

Where planning documents show work stations close to windows, a correspondingly wide strip can be taken as the work station area. The rest of the room less the ignored 0.5 m marginal strip is considered to be the surrounding area.

Uniformity required by ASR A3.4

Uniformity within the work station area should be 0.6, within the surrounding area 0.5.

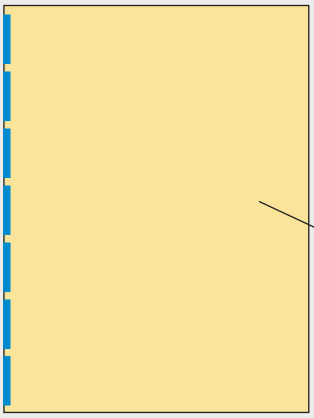


Office: Area of the room in which the arrangement of work stations and therefore the location of task areas are unknown at the design stage. Height: 0.75 m; 0.5 m marginal strip is ignored.

Office: Strips in which the approximate arrangement of work stations and therefore the location of task areas is known at the design stage. Height: 0.75 m; 0.5 m marginal strip is ignored.

06

[06] Definition of office areas



Office-like room: where it is known that work areas may extend to the boundaries of the room, the lighting area encompasses the whole room.

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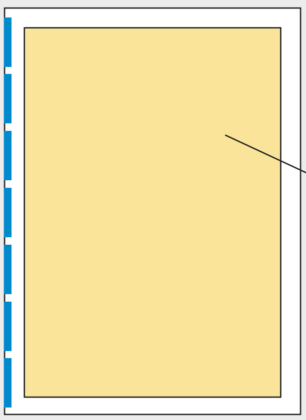
[07] Definition of office areas

a.3 Office-like room with possible arrangement of work stations extending to the boundaries of the room

Where it is known that working areas may extend to the boundaries of the room but the precise location of the work station areas is unknown, the whole room is taken to be the work area without deduction of any marginal zones.



School: room with flexible arrangement of student desks; a 0.5 m wide marginal strip is ignored.

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[08] Classrooms: maintained illuminance is 300 lux for primary and secondary schools, 500 lux for evening classes, adult education and lecture theatres.

b. Classroom with flexible arrangement of desks

Students' desks are often rearranged in a classroom, so lighting needs to cater for tasks performed anywhere in the room. A 0.5 m wide marginal strip can be ignored and deducted.

Uniformity is 0.60.

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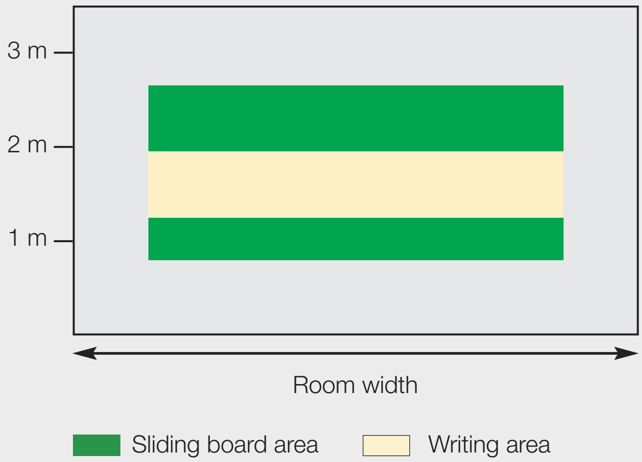
[09 + 10] Horizontal and vertical surfaces (boards, charts, posters) that may constitute task areas. In the case of boards, uniformity should be observed at writing height.



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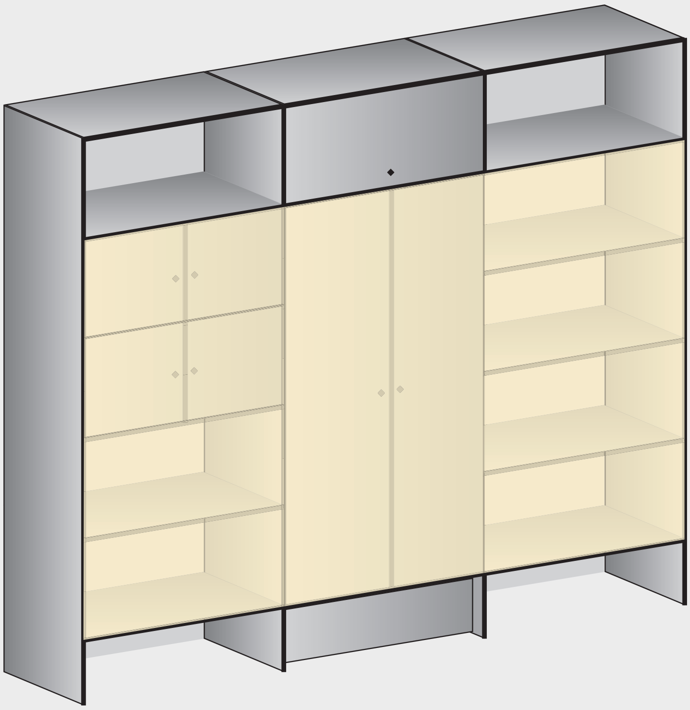
Vertical illuminance

Vertical illuminance in the main viewing direction should be $E_v > 100$ lx in classrooms with 300 lx illuminance and $E_v > 175$ lx in evening class rooms and lecture theatres with 500 lx illuminance. These requirements for compliance with ASR A3.4 also apply to walls with charts and posters. No requirements are specified for individual student desks.

500 lx vertical illuminance needs to be maintained over the whole surface of a chalkboard. A strip extending to each side of the board at a writing height of 1.2 – 1.8 m is used as a reference for 0.70 uniformity. Uniformity over the entire work surface should be 0.60 (cf. LiTG publication “Leitfaden zur Beleuchtung von Unterrichts- und Vortragsräumen” on classroom and lecture room lighting).



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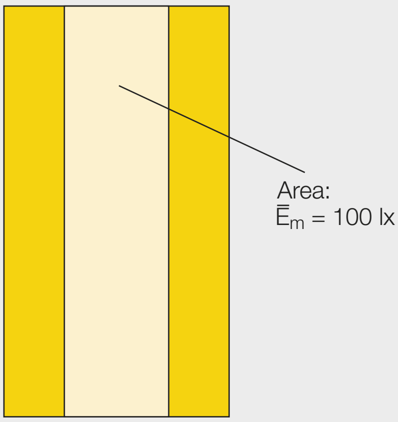
[11] Where visual tasks are performed mainly on a vertical plane, that plane is the task area.

c. Shelving systems and other vertical surfaces

Shelving systems and cabinets need to be regarded as vertical task areas if visual tasks need to be performed there over an extended period of time (e.g. ticket-issuing or bookkeeping). The vertical task area reference surface starts 0.5 m above floor level and, in the case of an office shelving system, ends 2.0 m above floor level.



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[12] Corridor: central strip as reference surface, surrounding area extends to walls

d. Corridor

In corridors, the entire area of the room in which traffic flows occur is regarded as the reference surface. For corridors up to 2.5 m wide, it is recommended – in line with DIN EN 1838 – that a central strip on the floor at least 1.0 m wide should be regarded as the reference surface and the rest of the space to the walls treated as surrounding area. In wider corridors, the central strip constituting the reference surface should be adjusted accordingly. Uniformity on the reference surface is 0.40. Walls require vertical illuminance $E_v > 50$ lx and a minimum uniformity of 0.10. Visual tasks here include doors, door handles and signs.

Maintained illuminance

For circulation areas and corridors with no vehicular traffic, ASR A3.4 requires 50 lx maintained illuminance and 0.6 uniformity; DIN EN 12464-1 stipulates 100 lx with 0.40 uniformity. The minimum values are comparable at 30 lx and 40 lx respectively.
100 lx maintained illuminance is recommended on the reference surface.

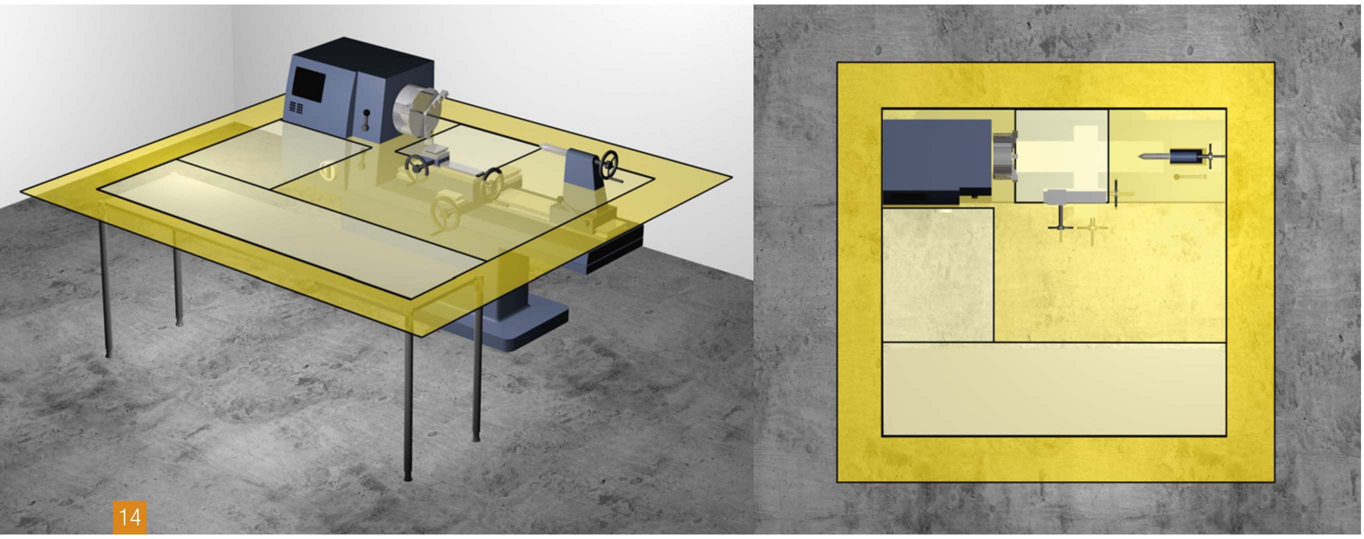
e. Single industrial work station

The visual tasks performed at an industrial work station are often numerous and diverse. They need to be defined individually in terms of location and size.

If the individual visual tasks are comparable, a work station area in which they are all performed can be defined.

The immediate surrounding area forms a band around the work station area at least 0.5 m wide. To ensure that enough light is available for all the workplaces in the bay, however, it is advisable to install a general lighting system that caters for the entire room. Where maintained illuminance > 500 lx is required, a task area lighting solution needs to be provided.

[13] Examples of work station task areas with differing requirements: area for turning and measuring moderately fine parts presenting vertical and horizontal visual tasks (1), area for studying drawings on vertical surfaces (2), area for checking workpiece measurements and depositing tools (3)



[14] Several task areas at a lathe considered as a single work station area (light and medium yellow). The surrounding area forms a strip around it at least 0.5 m wide (dark yellow).

[15] Industrial bay with zones for different activities

f. Industrial bay with zones for different activities
Industrial bays generally incorporate a number of task areas with diverse illuminance requirements. Where this is the case, it is recommended that, as a first step, a general hall lighting concept should be developed treating the whole hall – less a 0.5 m wide marginal strip along the walls – as a task area with the lowest requirements.

For the other task areas with different requirements, appropriate – preferably rectangular – task areas with their own surrounding areas should be defined and provided with the illuminances and uniformities required. *(see Fig. 15)*.

Task areas where maintained illuminance ≥ 750 lx is required should be provided with work surface lighting.

4. Calculation grid for the design, computation and verification of lighting installations

In principle, the grid required to determine average illuminance and uniformity depends on the size and shape of the reference surface considered. Reference surfaces are work station, surrounding and background areas, on the one hand, and activity or interior areas, on the other.

Consideration needs to be given here to the geometry of the lighting installation, the luminous intensity distribution of the luminaires, the degree of precision required and the photometric quantities to be evaluated.

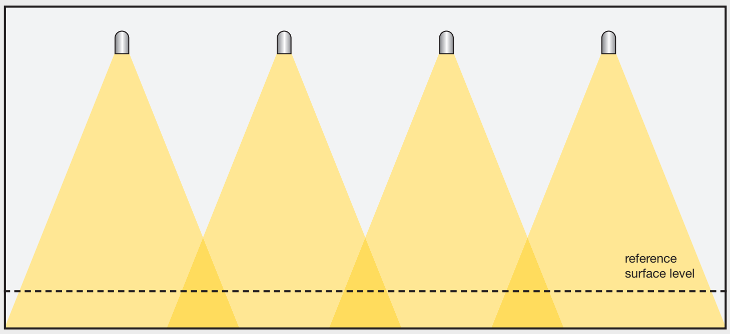
- The arrangement of luminaires and the arrangement of measurement points should not be identical.
- The spacing between measurement points needs to be less than the mounting height.
- In high bays, light beams should overlap at height and not just on the reference surface.

A 0.5m wide strip along the walls is excluded from the calculation area. This is unless task areas are located within the strip or extend into it.

For the precise definition of a calculation grid, see Appendix 3: "Calculation grid", page 36.

Size of grid recommended for rooms and areas

	Longest dimension of area or room	Grid size
Task area	approx. 1 m	0.2 m
Small rooms/ room zones	approx. 5 m	0.6 m
Medium-size rooms	approx. 10 m	1 m
Large rooms	approx. 50 m	3 m



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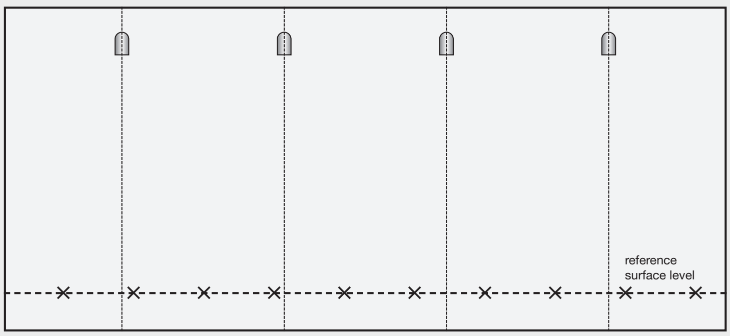

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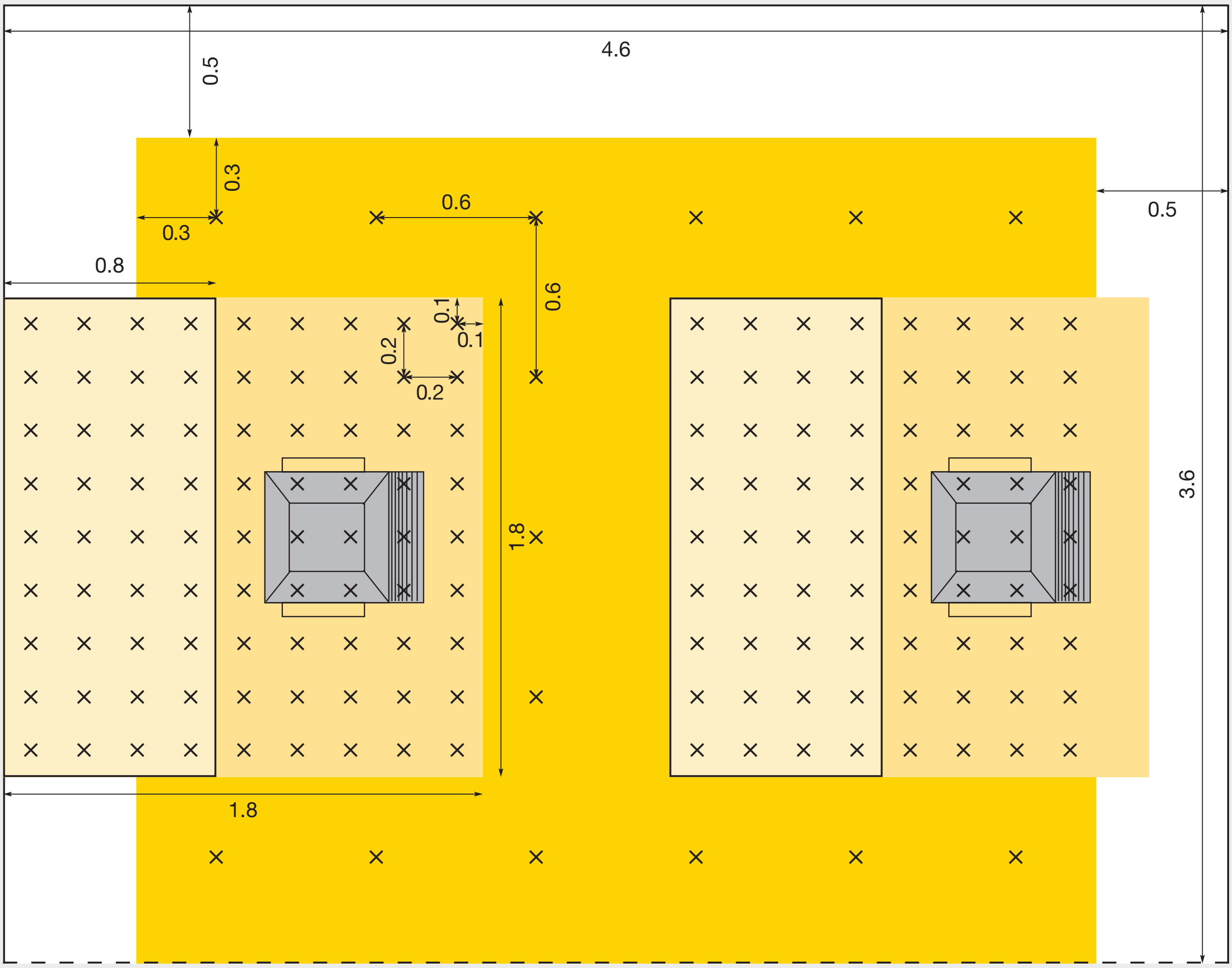
17

[16] Luminaires should be arranged so that their beams overlap at height. This is achieved by appropriate luminaire geometry and the right choice of beam characteristics.

[17] Measurement points should be selected so that their arrangement does not coincide with the arrangement of luminaires.



18

[18] Definition of calculation points in the surrounding area (dark yellow) and in the work station area (work space/desk: light yellow, movement space: medium yellow). A 0.5 m wide marginal strip is ignored unless the work space/desk projects into it.

Calculation points only for working surfaces

Where part of a work station area (work space + movement space) extends into the strip along the wall, calculation points need not be considered if the projecting area is movement space. However, if the surface extending into the marginal strip is work space (e.g. a desktop), calculation points need to be considered.

5. Illuminance for walls and ceilings

One new requirement in DIN EN 12464-1 is balanced luminance distribution. This is achieved by taking account of the luminance of all surfaces, which is determined by the reflectance of the surfaces and the illuminance on them. To avoid gloom, raise adaptation levels and enhance visual comfort, room surfaces should be bright, especially walls and ceilings.

Recommended reflectance for the major diffusely reflecting room surfaces:

- ceiling: 0.7 to 0.9
- walls: 0.5 to 0.8
- floor: 0.2 to 0.4

Maintained illuminance should be

- significantly higher than 50 lx on walls and
- over 30 lx on the ceiling.

In some enclosed spaces (e.g. offices, classrooms, hospitals, corridors and stairwells), it is recommended that maintained illuminance should be raised to 75 lx for walls and 50 lx for ceilings. Uniformity is required to be higher than 0.10 in each case. For bright, health-promoting rooms, illuminance targets should be significantly higher in high visual communication zones.

Bright rooms

ASR A3.4 sets out no values for illuminance on walls and ceilings. Like the revised standard, however, it manifestly attaches importance to bright interiors for certain forms of room use.

6. Lighting in the interior space

DIN EN 12464-1 stresses the importance of quality of lighting in the interior space. In addition to task lighting, lighting is required to illuminate the space occupied by persons. This light is needed to highlight objects, reveal textures and improve the appearance of persons in the room. The physical lighting conditions are expressed in terms of "mean cylindrical illuminance", "modelling" and "directional light".

6.1 Mean cylindrical illuminance $\bar{E}_z$

Maintained illuminance must be no lower than 50 lx. In places where good visual communication is crucial, e.g. in an office, meeting room or classroom, maintained illuminance should be raised to 150 lx.

This requirement needs to be met at 1.2 m above floor level for seated persons and 1.6 m above floor level for persons standing in activity and interior areas.

In both cases, uniformity is required to be higher than 0.10.

Care needs be taken to ensure that cylindrical illuminance requirements are met wherever faces are present.

Why is cylindrical illuminance a measure for illuminating faces?

Semi-cylindrical illuminance on the side of the face directed towards the observer would certainly be a more obvious choice. However, that would presuppose that viewing directions were known at the design stage and would also entail an unacceptable planning effort.
Studies have shown that when we look at faces, we tolerate very marked differences in vertical illuminance from different directions. In the case of typical workplace lighting installations with a uniform arrangement of luminaires on or parallel to the ceiling, the uniformity of the vertical illuminance values used to define cylindrical illuminance is a great deal higher than the uniformity tolerated. The use of cylindrical rather than semi-cylindrical illuminance is thus justified by the considerably lower planning effort required.

6.2 Modelling

Modelling is a good yardstick for 3D perception of persons and objects in a room. It expresses the balance between diffuse and directional light and is determined by the ratio of cylindrical illuminance to horizontal illuminance at a given point (normally 1.2 m above floor level). As a rough guide, a value between 0.30 and 0.60 is an indicator of good modelling: faces and bodies are not too dramatically shaded or sharply illuminated, nor are they cast in a flat, dull light.

Note: This ratio is referred to as "shadow effect" in the DIN 5035 series, where 0.3 is a minimum requirement.

6.3 Directional lighting of visual tasks

Directional light can emphasise details of a visual task. However, harsh disturbing shadows should be avoided.

DIN EN 12464-1 specifically points out the need to avoid multiple shadows, which can be caused by directional light from more than one point light source and can produce a confusing visual effect.

Vertical illuminance in the interior space

Mean vertical illuminance needs to be appropriate for the visual task and work performed. For some work environments, work stations or activities, ASR A3.4 requires a higher vertical illuminance of $E_v > 100$ lx (e.g. primary school classrooms) or $E_v > 175$ lx (e.g. career/technical classrooms, first aid rooms or writing and reading activities).
A proven ratio of vertical illuminance to horizontal illuminance is ≥ 1:3.

7. Limitation of glare

Glare is the sensation produced by excessively bright areas or excessively marked differences in luminance within an observer's field of view. Glare which causes direct impairment of vision is known as disability glare. Glare which is found disturbing, which impairs our sense of wellbeing, is known as discomfort glare.

7.1 Rating discomfort glare by the UGR method

The degree of discomfort glare caused by a lighting system can be determined by the UGR method *(see Appendix 4: "Rating interior lighting installations for glare", page 37 ff.)*. The UGR_L limit depends on the difficulty of the visual task and should not be exceeded. The following are examples of maximum limits:

Examples of maximum UGR_L limits

Technical drawing	≤ 16
Reading, writing, classrooms, computer work, inspections	≤ 19
Work in industry and craft workshops, reception	≤ 22
Rough work, staircases	≤ 25
Corridors	≤ 28

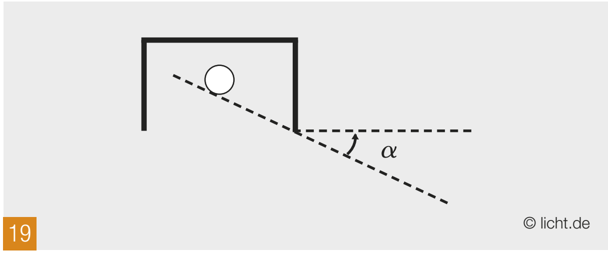
A lighting system should be appropriate for the relevant UGL category (e.g. "≤ 19"). UGR values can be ascertained by the tabular method. UGR tables are available in manufacturers' catalogues or databases.

For initial luminaire selection, it is advisable to use the tabular value of the reference room UGR_R (4H x 8H) based on a spacing-to-height ratio of 0.25 *(see page 39)*.

Individual UGR values in a lighting installation can be calculated by the formula method using CAD software *(see page 39)*. This may be useful for designing installations where glare is a critical factor but it does not indicate the standard of glare limitation of the installation as a whole.

7.2 Shielding

As excessively bright light sources in the field of vision can cause glare, lamps/light sources also need to be suitably shielded. For luminaires that are open from below or fitted with a clear enclosure, the shielding angle is defined as the angle between the horizontal and the line of sight below which the luminous parts of the lamp in the luminaire are directly visible.



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[19] Shielding angle α

The following table shows minimum shielding angles at specific lamp luminances.

Minimum shielding angles specified by DIN EN 12464-1

Lamp luminance in cd/m²	Minimum shielding angle
20,000 to < 50,000 e.g. fluorescent lamps (high output) and compact fluorescent lamps, LEDs	15°
50,000 to < 500,000 e.g. high-pressure discharge lamps and incandescent lamps with matt and inside-coated bulbs	20°
≥ 500,000 e.g. high-pressure discharge lamps and incandescent lamps with clear bulbs, high performance LEDs	30°

The minimum shielding angles for the lamp luminances shown need to be observed for all emission planes. They do not apply to luminaires with only a top-side light exit opening or to luminaires mounted below eye level.

7.3 Luminance limits for avoiding reflected glare

Special attention needs to be paid to avoiding glare caused by light reflecting from shiny surfaces (reflected glare). Reflections of excessively bright luminous parts of a luminaire can seriously interfere with work at a screen or keyboard, so care needs to be taken to arrange glare-critical luminaires so that no disturbing reflections are created.

In DIN EN 12464-1, luminance limits are specified for luminaires which could reflect along normal lines of sight from a screen inclined at up to 15°. Because display screen technology has advanced since the last edition of DIN EN 12464-1 was published in 2003, the limits are higher in the 2011 edition. Two limits are specified for ordinary office activities (positive polarity = dark characters on light background), depending on the luminance of the background:

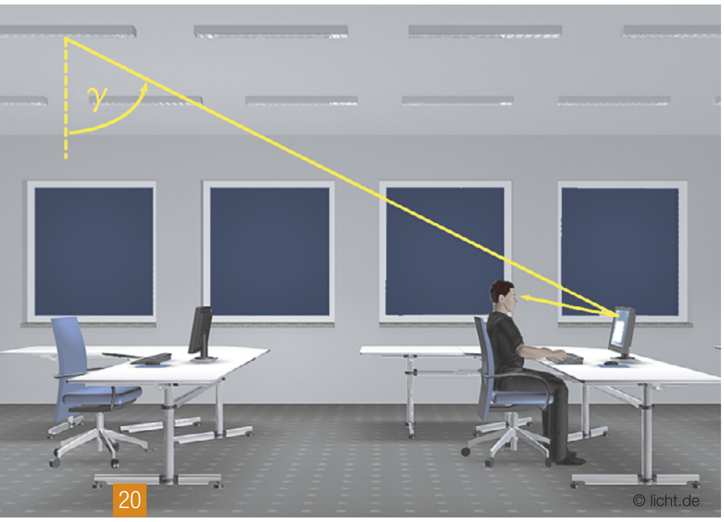
- For display screens where background luminance is $L \leq 200$ cd/m², luminaire luminance needs to be limited to a maximum value of 1,500 cd/m², whereas for screens where background luminance is $L > 200$ cd/m² luminaire luminances up to 3,000 cd/m² are permissible.
- For new flat screens, manufacturers generally indicate maximum adjustable background luminances $L > 200$ cd/m² but in practice the screens are mostly operated at ≤ 200 cd/m². What is more, the background luminance that is subsequently set is not known at the design stage. In such cases, the luminance of the luminaires used should not exceed 1,500 cd/m².
- Luminaires with luminance values up to a maximum of 3,000 cd/m² are allowed to be used only where it is ensured that screens have a background luminance $L > 200$ cd/m².
- Lower limits are set for more demanding visual tasks at a DSE (display screen equipment) work station (e.g. CAD).

The luminances specified must not be exceeded at elevation angles $\gamma \geq 65°$ from the downward vertical in any radiation plane.

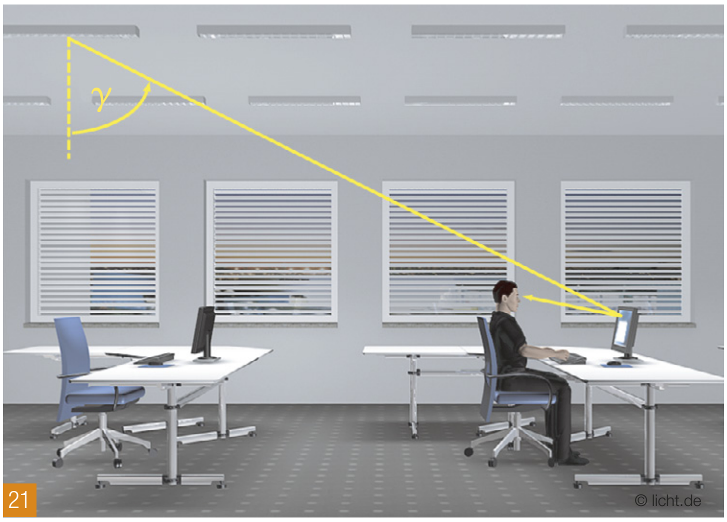
The values specified apply to flat-screen monitors with a good anti-glare – i.e. diffusely reflecting – finish, which are used at most office work stations today. Highly reflecting screens should not be used at constantly manned work stations.

The requirements set out in DIN EN 12464-1 do not apply to notebooks, laptops, tablet PCs or similar devices. Because they can be set up at any angle in any direction, disturbing reflections can be avoided by adjusting the position of the screen.

These requirements also meet the general stipulations set out in ASR A3.4 for the avoidance of reflected glare.



[20] For displays screens with background luminance $L \leq 200$ cd/m² (typical for offices with normal (average) daylight supply and for ordinary use of flat screens), luminaire luminances up to 1,500 cd/m² are permissible.



[21] For display screens with background luminance $L > 200$ cd/m² (typical for offices with good and very good daylight supply and for flat screens adjusted to the bright room situation), luminaire luminances up to 3,000 cd/m² are permissible.

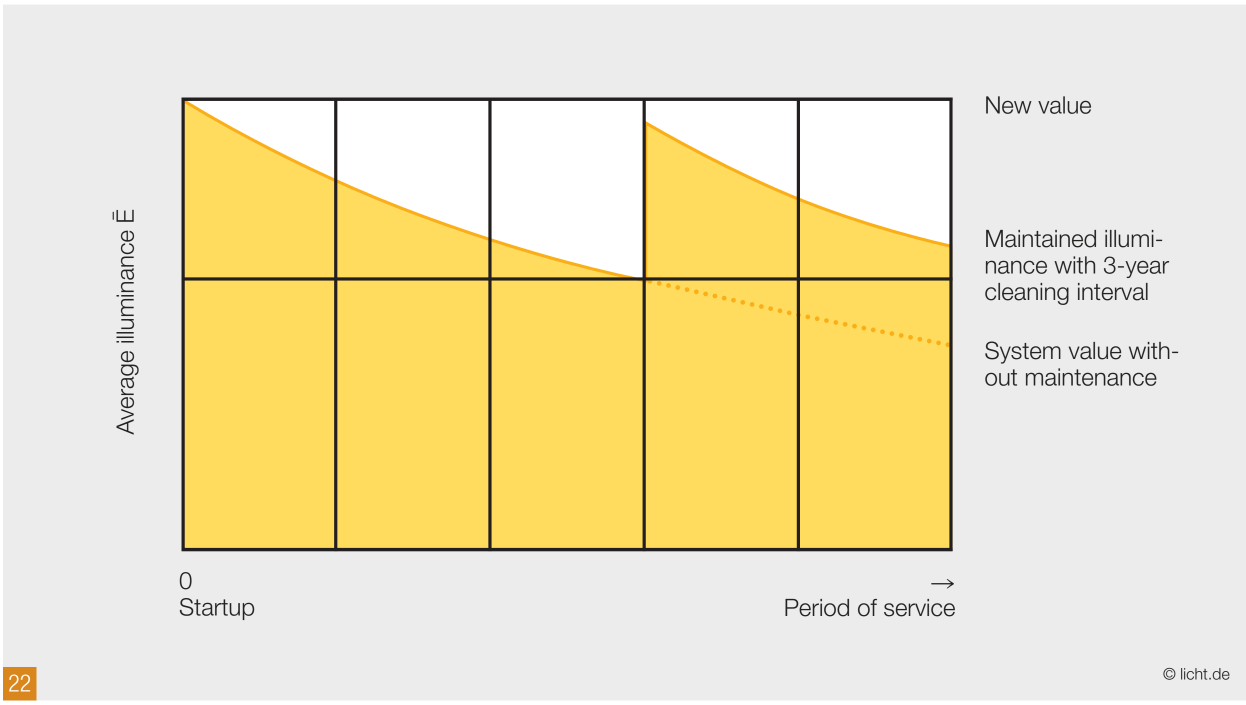
8. Lighting installation maintenance

With increasing length of service, the luminous flux delivered by a lighting system decreases as lamps and luminaires age and accumulate dirt. The anticipated decline of luminous flux depends on the choice of lamps, luminaires and operating gear, on the surfaces in the room and on the operating and environmental conditions to which the lighting installation is exposed.

For compliance with ASR A3.4, faults such as lamp failure or loss of illuminance, e.g. due to ageing or soiling of luminaires, need to be rectified immediately. Accordingly, maintenance of the lighting installation needs to be guaranteed.

To ensure that a specific lighting level – expressed by maintained illuminance – is reached for a reasonable period of time, an appropriate maintenance factor needs to be applied by the lighting designer to take account of this decrease in system luminous flux.

The maintenance factor (MF) of a lighting installation is the ratio of the luminous flux at the time of maintenance to the original luminous flux when the system is installed.



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[22] Illuminance during the period of service of a lighting installation – in this case with maintenance carried out every three years

8.1 Documenting maintenance factors

The designer needs to

- state the maintenance factor MF and list all assumptions made in determining its value
- specify lighting equipment suitable for the application environment and
- prepare a maintenance schedule, which should specify the frequency of lamp replacement, luminaire and room cleaning intervals and the cleaning techniques used.

The maintenance factor in the example on the right is 0.67 (values from CIE publication 97) subject to the following conditions: lamps are replaced in groups approximately every 16,000 operating hours, luminaires are cleaned every three years and room surfaces are cleaned every six years.

Example of maintenance factor documentation

Project:	office building, Frankfurt	
Room:	2-person office, room no. 0214	
Processed by:	Mr. Schulz	
Date:	02.03.2012 / 11:47:25	
Luminaire:	recessed luminaire	
Description:	luminaire xyz	
Article number:	123456789	
Luminaire type:	enclosed IP2X	
Cleaning interval in years:	3.0 (clean environment)	
Luminaire maintenance factor LMF:		0.79
Lamp:	fluorescent lamp, Ø 16mm	
Description:	T16 High Output	
Watt rating:	49 W	
Lamp replacement:	group/individual replacement of defective lamps	
Operating gear:	EB	
Lamp maintenance in years:	6.0	
Operating hours per lamp/year:	2,750 h	
Lamp lumen maintenance factor LLMF:		0.90
Lamp survival factor LSF:		1.00
Room:		
Length:	8 m	
Width:	6 m	
Height:	3 m	
Environment:	clean	
Room cleaning interval in years:	6.0	
Type of lighting:	direct	
Room maintenance factor RMF:		0.94
Maintenance factor MF:		**0,67**

8.2 Determining maintenance factors

The maintenance factor (MF) is a multiple of factors and is determined as follows:

MF = LLMF x LSF x LMF x RMF

where LLMF is the lamp lumen maintenance factor, LSF the lamp survival factor, LMF the luminaire maintenance factor and RMF the room maintenance factor. *(see Appendix 5: "Notes on maintenance factors", page 40)*

In many cases, a lamp survival factor (LSF) = 1 can be assumed because the failure of individual lamps leads to unacceptable falls in lighting level, which is why individual lamp replacement is required

Individual maintenance factor values can be obtained from manufacturers or found in manufacturer-independent standard average value curves (e.g. ZVEI publication: "Life behaviour of discharge lamps for general lighting", 2005) or in CIE publication 97 (2005).

Maintenance factors and conditions

Where one or more of the following – potentially inter-impacting – conditions applies, maintenance factors can generally be increased.

0,80

- Use of lamps subject to little light depreciation (depending on burning life), e.g. fluorescent lamps
- Use of luminaires with little tendency to collect dust
- Use of operating gear that lengthens lamp life (e.g. EB)
- Short periods of service per year
- Low switching frequency
- Short cleaning and/or maintenance intervals, individual and group lamp replacement
- Low exposure to dust in the atmosphere
- Low tendency to collect dust and/or for reflecting surfaces to become discoloured

0,67

- Use of lamps subject to marked light depreciation (depending on burning life), e.g. metal halide lamps
- Use of luminaires with tendency to collect dust
- Long periods of service per year
- High switching frequency per day
- Long cleaning and/or maintenance intervals (e.g. because of difficult access) only group lamp replacement
- High exposure to dust in the atmosphere
- Tendency to collect dust and/or for reflecting surfaces to become discoloured

0,50

Where one or more of the above – potentially inter-impacting – conditions applies, maintenance factors generally need to be lowered.

8.3 Decision paths for choosing maintenance factors

The above multiplication used to derive a maintenance factor from its individual components offers the lighting designer lots of opportunities to optimise lighting system maintenance intervals – and thus lighting system investment and operating costs – through the use of suitable lamps, luminaires and operating gear.

Many lamps have a long life. It would be unrealistic to assume that lamps need to be replaced before the end of their rated economic life.

Lamp life behaviour differs widely. For example:

- Compact fluorescent lamp: luminous flux declines to 85% after 10,000 hours
- T 16 fluorescent lamp: luminous flux declines to 89% after 24,000 hours
- Metal halide lamp (HCI-T 150W): luminous flux declines to 69% after 12,000 hours
- LED, e.g. for an LED module L70 = 50,000 hours (70% of the initial luminous flux is still available after 50,000 operating hours).

Frequent cleaning of lighting installations is also rarely a reality.

It is therefore advisable to assume longer maintenance intervals and choose a reference maintenance factor that ensures lighting installation operation stays above specified maintained values even after years of use with long-life lamps.

To prepare optimal maintenance schedules on the basis of manufacturers' current data and furnish documentation for a lighting design, it is advisable to use manufacturers programs or lighting design software such as Dialux and Relux.

8.4 Factors influencing the determination of maintenance factors

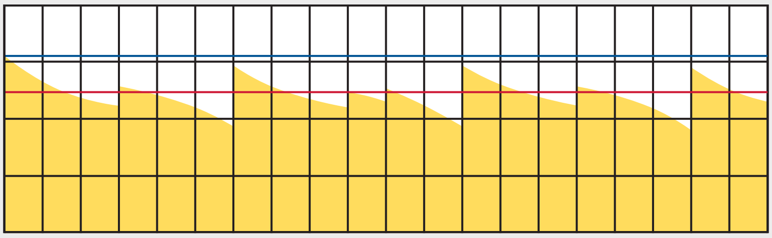
The maintenance factor can be optimised in two ways:
- Short maintenance intervals and a low initial illuminance value
- Longer maintenance intervals and, as a result, a higher initial illuminance value

The maintenance factor has a major impact on energy efficiency. The assumptions made in establishing the maintenance factor need to be optimised to produce a higher value without giving rise to excessively high costs for frequent maintenance.

The following charts show how the individual parameters impact on maintenance factors, maintenance intervals and observance of maintained illuminance in relation to overall costs.

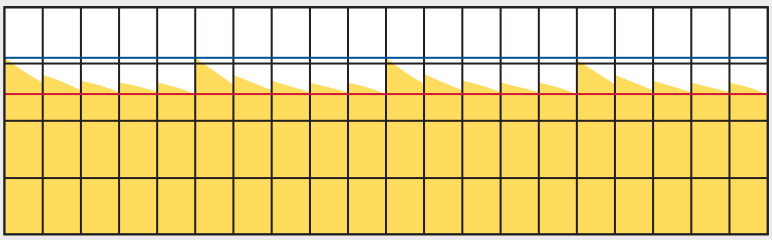
[23] Three examples showing the latitude available to the designer determining a maintenance factor.

Maintenance factor and total cost



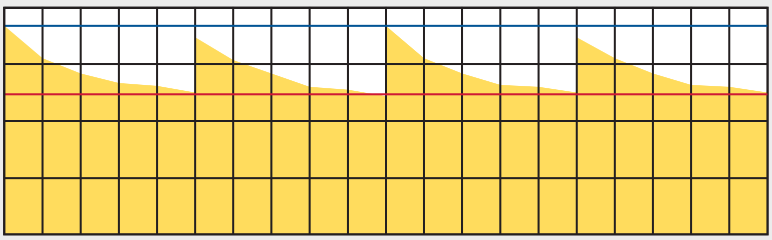
Lighting level not observed

Maintenance factor 0.80 (100 luminaires)
Luminaire cleaning every 3 years
Room maintenance every 10 years
Lamp replacement: group every 6 years
Total cost: -10% compared to base reference but with a lighting level short-fall of more than 20%



Lighting level observed but maintenance cycles idealised

Maintenance factor 0.80 (100 luminaires)
Luminaire cleaning every year
Room maintenance every 5 years
Lamp replacement: individual and group every 5 years
Total cost: 100% (base reference)



Lighting level observed and maintenance cycles realistic

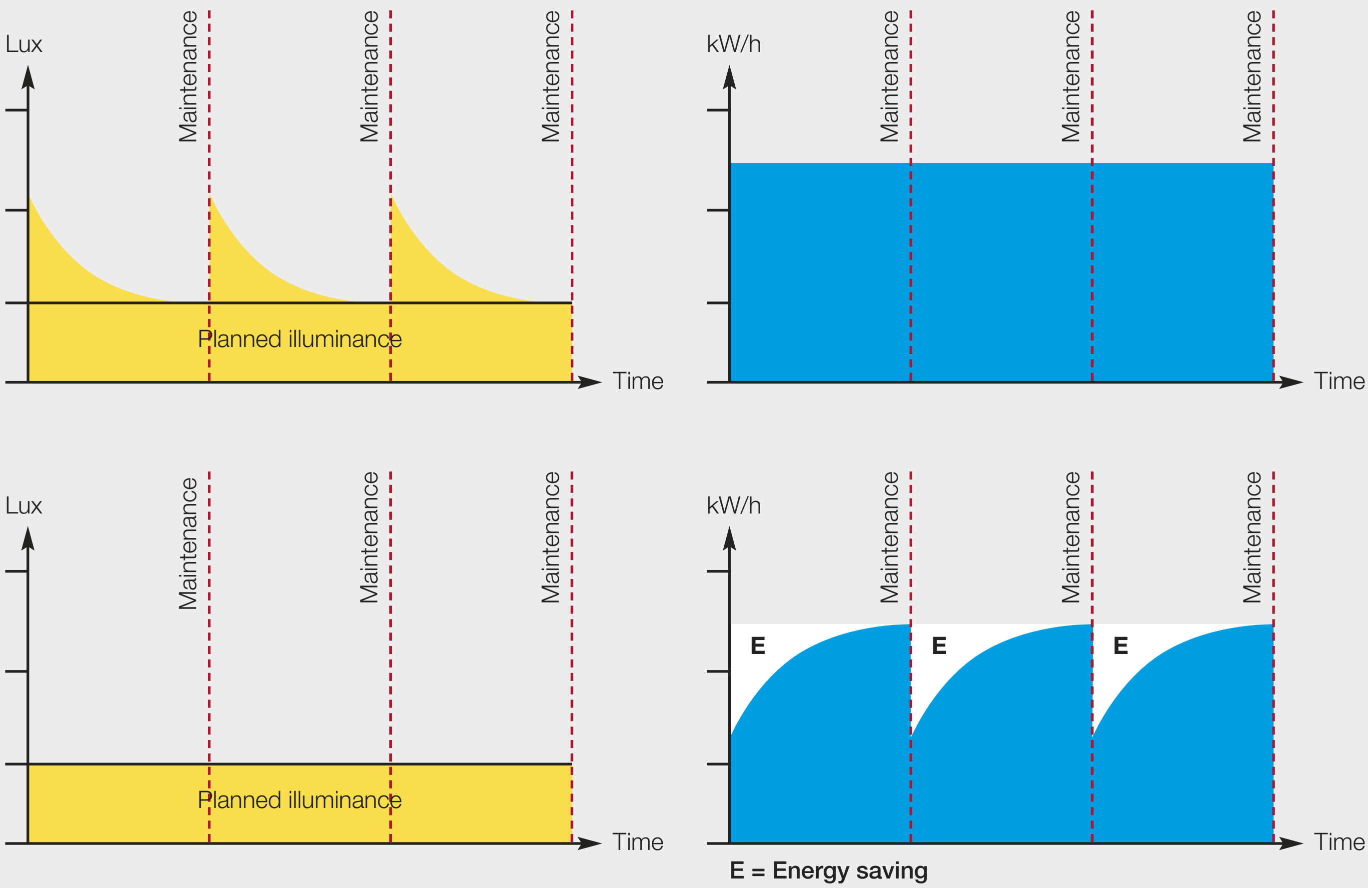
Maintenance factor 0.67 (120 luminaires)
Luminaire cleaning every 5 years
Room maintenance every 10 years
Lamp replacement: group every 5 years
Total cost: identical to base reference

General conditions: in each case luminaire type C (CIE 97) | direct/indirect | Very clean environment | 2,800 h annual operation | 12 ct/kWh (incl. 3% p.a. inflation) | Exemplary luminaire price € 150 | Luminaires with 2 x T16 54W EB | Payroll costs for maintenance € 50/h | Luminaire cleaning 15 min/luminaire | Lamp replacement 10 min/luminaire | Room maintenance € 5/m² | Room area 20 m x 40 m | Reflectances 70/50/20

23

Where installations are designed for a high initial value and long maintenance intervals, modern control and regulation technology enables illuminance to be kept constant at around the maintained illuminance mark. This is also pointed out in the statutory occupational accident insurers' office lighting guide BGI 856 (2008).

[24] Modern control and regulation technology helps keep illuminance constant at around the maintained illuminance mark.



Top: When the installation is new and each time maintenance is carried out, higher illuminance is briefly achieved while energy consumption remains constant.
Bottom: If illuminance is kept constant at a level just above the planned value, energy savings can be made.

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8.5 Maintenance factors

For rough projections or where detailed information is not available, one of the following values can initially be selected:

Maintenance factor	New-value factor	Example
0.80	1.25	very clean room, low-use installations
0.67	1.50	clean room, 3-year maintenance cycle
0.57	1.75	interior and exterior lighting, normal environmental pollution load. 3-year maintenance cycle
0.50	2.00	interior and exterior lighting, dirty environment

Use of the above values does not release designers from their documentation obligation.

A maintenance factor of 0.67 is recommended for comparing lighting designs without maintenance.

8.6 Examples of the determination of maintenance factors

The following maintenance factors are derived for two applications. The maintenance cycles assumed are realistic. The figures are in line with CIE 97 and data provided by lamp and luminaire manufacturers.

Example 1: Logistics centre

- Luminaire types:
  - high bay downlighter with high-pressure metal halide lamp
  - continuous row system with fluorescent lamps
  - LED panel luminaire: L_{70} = 75,000 h
- 4,000 operating hours a year
- Low environmental pollution load
- Reflectances: 50/30/20 (ceiling, walls, floor)

Replacement and cleaning intervals

Solution a

High bay downlighter with high-pressure metal halide lamp

- group lamp replacement and luminaire cleaning every 2 years
- individual replacement of defective lamps

Solution b

High bay downlighter with high-pressure metal halide lamp

- group lamp replacement and luminaire cleaning every 2 years

Solution c

Continuous row system with fluorescent lamps

- luminaire cleaning every 2 years
- group lamp replacement every 4 years

Solution d

LED panel luminaire (L_{70} = 75,000 h)

- luminaire cleaning every 2 years
- group PCB and driver replacement every 16 years
- individual replacement of defective circuit boards and drivers

Solution e

LED panel luminaire (L_{70} = 75,000 h)

- luminaire cleaning every 2 years
- group PCB and driver replacement every 16 years

		Solution a	**Solution b**	**Solution c**	**Solution d**	**Solution e**
		High bay downlighter with HPI* Luminaire cleaning & group lamp replacement every 2 years (8,000 h) Individual replacement of defective lamps	High bay downlighter with HPI* Luminaire cleaning & group lamp replacement every 2 years (8,000 h)	Continuous row system with TL** Luminaire cleaning every 2 years (8,000 h) & group lamp replacement every 4 years (16,000 h)	Panel luminaire with LED* Luminaire cleaning every 2 years (8,000 h) & PCB and driver replacement every 16 years (64,000 h) Individual replacement of defective PCBs or drivers	Panel luminaire with LED* Luminaire cleaning every 2 years (8,000 h) & PCB and driver replacement every 16 years (64,000 h)
LLMF	Lamp lumen maintenance factor	0.73	0.73	0.90	0.79	0.79
LSF	Lamp survival factor	1.00	0.87	0.95	1.00	0.98
LMF	Luminaire maintenance factor	0.94*	0.94*	0.86**	0.94*	0.94*
RMF	Room maintenance factor	0.95	0.95	0.95	0.95	0.95
MF	**Maintenance factor**	**0.65**	**0.57**	**0.70**	**0.71**	**0.69**

* enclosed luminaire ** open luminaire

Example 2: Office lighting

- Luminaire types:
  - recessed luminaires with fluorescent lamps
  - recessed luminaires with LEDs: L_{70} = 50,000 h
- 2,750 operating hours a year
- Clean environment
- Reflectances: 70/50/20 (C/W/F)

Replacement and cleaning intervals

Solution a

Recessed luminaires with fluorescent lamps

- group lamp replacement every 6 years
- individual replacement of defective light sources

Solution b

Recessed luminaire with LEDs (L_{70} = 50.000 h)

- group PCB and driver replacement every 15 years
- individual replacement of defective PCBs

		Solution a	**Solution b**
		Recessed luminaires with T16 fluorescent lamps group lamp replacement & luminaire cleaning every 6 years (16,500 h) Individual replacement of defective light sources	Recessed luminaires with LED & enclosed optics PCB & driver replacement every 15 years (41,000 h) Individual replacement of defective PCBs or drivers
LLMF	Lamp lumen maintenance factor	0.90	0.80
LSF	Lamp survival factor	1.00	1.00
LMF	Luminaire maintenance factor	0.86**	0.92*
RMF	Room maintenance factor	0.94	0.94
MF	**Maintenance factor**	**0.73**	**0.69**

* enclosed luminaire ** ** open luminaire

9. Appendices

9.1 Appendix 1: Changes in DIN EN 12464-1:2011 compared to DIN 12464-1:2003

The main technical changes are:

- the importance of daylight has been taken into account: requirements for lighting are applicable regardless of whether artificial lighting, daylight or a combination of the two is used;
- specification of a minimum illuminance on walls and ceilings;
- specification of cylindrical illuminance and detailed information on modelling;
- uniformity of illuminance is assigned to tasks and activities;
- definition of "background area" with lighting specification for this area;
- definition of an illuminance grid in accordance with DIN EN 12464-2;
- new luminance limits for luminaires used with flat panel displays (display screen equipment (DSE) as defined in ISO 9241-307).

Differences in values

Maintained illuminance values $\bar{E}_m$ have been changed in a small number of cases; a few new interior areas, task areas and activity areas have been added.

Lower $\bar{E}_m$

- Stairs, escalators, travelators from 150 lx to 100 lx (5.1.2)
- Health care premises: corridors, during the day, from 200 lx to 100 lx (5.37.2)

Higher $\bar{E}_m$

- Eye examination rooms: general lighting from 300 lx to 500 lx (5.41.1)
- Ear examination rooms: general lighting from 300 lx to 500 lx (5.42.1)

Colour rendering requirements have been adjusted in a few cases. $R_a > 80$ is specified as a basic minimum at constantly manned work stations.

Additions:

- Elevators, lifts (5.1.3)
- Storage rack face (5.5.4)
- Health care premises:
  - Corridors: cleaning (5.37.3): 100 lx
  - Corridors with multi-purpose use (5.37.5): 200 lx
  - Elevators, lifts for persons and visitors (5.37.7): 100 lx
  - Service lifts (5.37.8): 200 lx
- Railway installations:
  - Fully enclosed platforms, small number of passengers (5.53.1): 100 lx
  - Fully enclosed platforms, large number of passengers (5.53.2): 200 lx
  - Passenger subways (underpasses), large number of passengers (5.53.4): 100 lx
  - Entrance halls, station halls (5.53.8): 200 lx
  - Switch and plant rooms (5.53.9): 200 lx
  - Access tunnels (5.53.10): 50 lx
  - Maintenance and servicing sheds (5.53.11): 300 lx

9.2 Appendix 2: Differences between DIN EN 12464-1:2011 and ASR

Values in DIN EN 12464-1

Ref. no.	Type of area	$\bar{E}_m$	R_a
Traffic zones inside buildings			
5.1.1	Circulation areas and corridors	100	40
5.1.1	Circulation areas and corridors	100	40
	– no specification –		
	– no specification –		
General areas inside buildings – Store rooms, cold stores			
5.4.1	Store and stockrooms	100	60
5.4.2	Dispatch packing handling areas	300	60
	– no specification –		
General areas inside buildings – Rest, sanitation and first aid rooms			
5.2.2	Rest rooms	100	80
General areas inside buildings – Control rooms			
5.3.1	Plant rooms	200	60
Industrial activities and crafts – Cement, cement goods, concrete, bricks			
5.8.1	Drying	50	20
Industrial activities and crafts – Ceramics, tiles, glass, glassware			
5.9.1	Drying	50	20
	– no specification –		
Industrial activities and crafts – Chemical, plastics and rubber industry			
5.10.1	Remote-operated processing installations	50	20
Industrial activities and crafts – Foundries and metal casting			
5.13.3	Sand preparation	200	80
5.13.8	Machine moulding	200	80
5.13.4	Dressing room	200	80
5.13.6	Casting bay	200	80
5.13.7	Shake out areas	200	80
5.13.9	Hand and core moulding	300	80
5.13.10	Die casting	300	80
Industrial activities and crafts – Metal working and processing			
5.18.1	Open die forging	200	80
5.18.2	Drop forging	300	80
5.18.3	Welding	300	80
5.18.4	Rough and average machining: tolerances ≥ 0,1 mm	300	80
5.18.5	Precision machining; grinding: tolerances < 0,1 mm	500	80
5.18.6	Scribing; inspection	750	80
5.18.7	Wire and pipe drawing	300	80
5.18.8	Plate machining	200	80
5.18.9	Sheet metalwork	300	80
5.18.10	Tool making, cutting equipment manufacture	750	80
	– no specification –		
Industrial activities and crafts – Power stations			
5.20.1	Fuel supply plant	50	20
	– no specification –		
Industrial activities and crafts – Rolling mills, iron and steel works			
5.22.1	Production plants without manual operation	50	20
5.22.3	Production plants with manual operation	200	80
Industrial activities and crafts – Wood working and processing			
5.25.2	Steam pits	150	40
5.25.3	Saw frame	300	60
Places of public assembly – General areas			
5.28.1	Entrance halls	100	80
Places of public assembly – Theatres, concert halls, cinemas, places for entertainment			
5.30.2	Dressing rooms	300	90
Places of public assembly – Libraries			
5.33.1	Bookshelves	200	80
Educational premises – Educational buildings			
5.36.4	Black, green and white boards	500	80
Health care premises – Rooms for general use			
5.37.2	Corridors: during the day	100	80
Health care premises – Wards, maternity wards			
5.39.1	General lighting	100	80
Health care premises – Intensive care unit			
5.47.4	Night watch	20	90
	– no specification –		
	– no specification –		

Values in ASR A3.4

■ Value lowered ■ Value raised ■ Other difference

Ref. no.	Type of area	$\bar{E}_m$	R_a
Circulation routes			
1.1	Circulation areas and corridors with no vehicular traffic	**50**	40
1.2	Circulation areas and corridors with vehicular traffic	**150**	40
1.6	Vehicle entrances of industrial buildings during the day	400	40
1.6	Vehicle entrances of industrial buildings at night	50	40
Storage facilities			
2.2	Store rooms for identical or large stored goods	**50**	60
2.3	Store rooms with searches for diverse stored goods	**100**	60
2.4	Store rooms where reading tasks are performed	200	60
General areas, activities and tasks			
3.2	Rest, waiting, recreation rooms	**200**	80
3.6	Building service equipment, switch gear rooms	200	**80**
Cement, concrete and brick industry			
7.1	Drying	50	**40**
Ceramics, tiles, glass, glassware, optician			
8.1	Drying	50	**40**
8.6	Optician's workshop	1500	90
Chemical industry, plastics and rubber industry			
9.1	Remote-operated processing installations	50	**40**
Metal working and processing, foundries and metal casting			
16.1	Sand preparation and other tasks	200	**60**
16.1	Machine moulding	200	**60**
16.1	Casting bays	200	**60**
16.1	Shake out areas	200	**60**
16.1	Dressing room	200	**60**
16.2	Hand and core moulding	300	**60**
16.2	Die casting	300	**60**
Metal working and processing, foundries and metal casting			
16.4	Open die forging	200	**60**
16.5	Drop forging	**200**	**60**
16.6	Welding	300	**60**
16.7	Rough and average machining: tolerances ≥ 0,1 mm	300	**60**
16.8	Precision machining; grinding: tolerances < 0,1 mm	**300**	**60**
16.9	Scribing, inspection	750	**60**
16.10	Wire and pipe drawing	300	**60**
16.11	Plate machining	200	**60**
16.12	Sheet metalwork	300	**60**
16.13	Tool making, cutting equipment manufacture	750	**60**
16.18	Motor vehicle repair shops and inspection stations	300	80
Power stations			
18.1	Fuel supply plant	50	**40**
18.5	Outdoor substations	20	40
Rolling mills, iron and steel works			
20.1	Production plants without manual operation	50	**40**
20.2	Production plants with manual operation	200	**40**
Wood working and processing			
23.2	Steam pits	**100**	40
23.3	Saw frame	**200**	60
General areas, activities and tasks			
3.11	Entrance halls	**200**	80
General areas, activities and tasks			
3.4	Dressing rooms	**200**	80
Libraries			
26.1	Bookshelves	**200 vertikal**	80
Educational buildings, nursery schools, pre-schools			
27.4	Boards	**500 vertikal**	80
Health care premises			
28.1	Corridors: during the day	**200**	80
Health care premises			
28.3	General lighting	**200**	80
Health care premises			
28.8	Monitoring of patients at night	**50**	90
28.12	Preparation of instruments	500	80
28.13	Health care laboratories	500	90

9.3 Appendix 3: Calculation grid

Experience has shown that the following grid size p should not be exceeded:

$$p = 0.2 \times 5^{\log_{10} d}$$

where:
p is the grid size and d the relevant dimension of the reference surface. The number of points is then given by the next whole number of the ratio d to p.

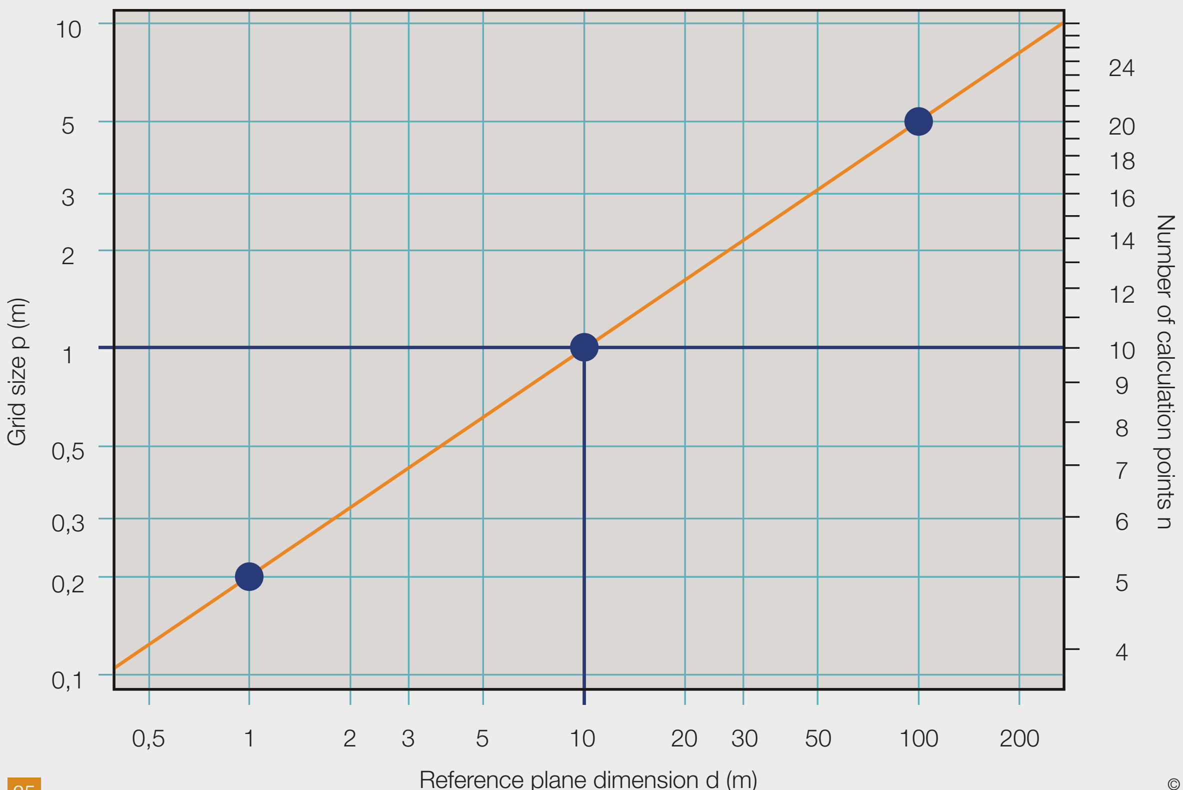
Rectangular reference surfaces are subdivided into smaller, roughly square rectangles with the calculation points at their centre. The arithmetic mean of all the calculation points is the average illuminance. Where the reference surface has a length-to-width ratio between 0.5 and 2.0, the grid size p and therefore the number of points can be determined on the basis of the longer dimension d of the reference area. In all other cases, the shorter dimension needs to be taken as the basis for establishing the spacing between grid points.

For non-rectangular reference surfaces, i.e. surfaces restricted by irregular polygons, grid size can be determined analogously using an appropriately dimensioned circumscribing rectangle. Arithmetic means and uniformities are then established taking only the calculation points within the restricting polygons of the reference surface.

For ribbon-like reference surfaces, which normally result from the surrounding areas evaluated, the dimension of the ribbon at its widest point should be taken as the basis for determining grid size. However, the grid size thus established must be no greater than half the dimension of the ribbon at its narrowest point if that is 0.5 m or more. Arithmetic means and uniformities are again determined taking only the calculation points within the ribbon.

[25] Grid size as a function of reference plane dimensions

Grid point spacing according to DIN EN 12464-1



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9.4 Appendix 4: Rating interior lighting installations for glare

Direct glare caused by luminaires in an indoor lighting system can be rated using the CIE Unified Glare Rating (UGR) method. This method is based on the formula:

$$UGR = 8 \log_{10} \left(\frac{0{,}25}{L_b} \Sigma \frac{L^2 \omega}{p^2} \right)$$

where:

L_b the background luminance in cd/m², calculated as E_{ind}/π, in which E_{ind} is the vertical indirect illuminance at the observer's eye,

L the average luminance in cd/m² of the luminous parts of the luminaire in the direction of the observer,

ω the solid angle in sr of the luminous parts of the luminaire visible from the vantage of the observer,

p the Guth position index for each individual luminaire.

Use of the UGR method is restricted to direct luminaires and direct/indirect luminaires with an indirect component up to 65 percent. In the case of luminaires with an indirect component > 65 percent, the UGR method produces unduly favourable ratings. Generally speaking, however, glare can be largely ruled out in the case of these luminaires because of the very low glare potential of the direct component.

According to CIE Publication 117, the UGR method can no longer be used for large light sources (solid angle > 1 sr) or small light sources (solid angle < 0.0003 sr).

Large light sources can be individual luminaires with luminous surfaces > 1,5 m², luminous ceilings with at least 15 percent luminous panelling or uniformly illuminated ceilings.

As the dazzling effect of large light sources depends to only a small extent on their position index, solid angle or background luminance, the glare caused by large light sources can be fairly approximated on the basis of luminance and limited by defining a maximum permissible value. In DIN 5035 Part 1, the maximum permissible luminance was set at 500 cd/m2. In LiTG Publication 20 on the UGR method, the limit recommended for limiting glare to a UGR of 19 is 350 cd/m² for large rooms and 750 cd/m² for small rooms.

Small light sources visible below a solid angle < 0,0003 sr are generally found in the following situations:

a. in low interiors (room height h < 3 m, e.g. office lighting systems). Downlights, for example, can occupy small solid angles here if they are a fairly long way from the observer.
b. in high halls (e.g. sports and industrial hall lighting systems). High-bay reflector luminaires, for example, are visible to the observer at small solid angles here because of their high mounting height.

In both cases, glare due to light sources < 0,0003 sr cannot be ruled out. Drawing on field study findings, LiTG Publication 20 therefore recommends that the lower solid angle limit should be abolished to avoid situations where glare fails to be anticipated because disturbing luminaires are below the solid angle limit and are therefore disregarded.

Rating by the tabular method
According to the standard, the degree of direct glare caused by a lighting system can be determined using the UGR tabular method.

Here, the system concerned is compared with a standard table listing UGR values for 19 standard rooms and various reflectance combinations for the selected luminaire. The computations for the 19 standard rooms are based on the assumption that the observers – positioned at the midpoint of each wall – observe the luminaires along and across their lines of sight along the room axes. The luminaires are mounted in a regular grid on the luminaire plane, the midpoints of the luminaires set at a distance 0.25 times the distance H between the luminaire plane and the height of the observer's eye and the midpoints of the luminaires closest to the walls set half as far from the wall as the luminaire midpoints from each other.

When selecting suitable luminaires, care must be taken to ensure that only tables with the same spacing-to-height ratio and the same lamp luminous flux are compared.

A “Table of corrected standardised glare ratings” is shown on page 39.

Rating in the reference room
If not all UGR tables are available or if dimensions or reflectances are unknown at the design stage, glare can be rated using the UGR value for the **reference room**.

The reference room is a medium-sized room measuring 4H x 8H with ceiling, wall and floor reflectances of 0.7, 0.5 and 0.2 respectively. The ranking resulting from comparison of different lighting systems is generally maintained provided the UGR values compared were computed for the same luminaire midpoint spacing and the same lamp luminous flux. At all events, glare rating must be based on the installation values of the lighting systems and the rated values of the lamps used.

Whichever method is used, the UGR values thus established must not exceed the UGR limits for interiors, tasks and activities stated in the “Schedule of lighting requirements” tables contained in the standard.

Table of corrected standardised glare ratings (UGR)

		Luminaire spacing/mounting height above observer's eye a/h = 0.25 Reflectances									
Ceiling		0.70	0.70	0.50	0.50	0.30	0.70	0.70	0.50	0.50	0.30
Walls		0.50	0.30	0.50	0.30	0.30	0.50	0.30	0.50	0.30	0.30
Floor		0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Dimensions		**Corrected glare ratings – luminous flux 5.200 lm**									
X	**Y**	**Across line of sight**					**Along line of sight**				
2H	2H	16.4	18.0	16.8	18.3	18.6	17.4	19.0	17.7	19.2	19.5
	3H	16.3	17.7	16.6	18.0	18.3	17.2	18.6	17.6	19.0	19.3
	4H	16.2	17.5	16.6	17.9	18.2	17.2	18.5	17.5	18.8	19.2
	6H	16.2	17.4	16.6	17.7	18.1	17.1	18.3	17.5	18.7	19.0
	8H	16.2	17.3	16.6	17.6	18.0	17.1	18.2	17.5	18.6	18.9
	12H	16.1	17.2	16.5	17.5	17.9	17.1	18.1	17.5	18.5	18.9
4H	2H	16.4	17.7	16.8	18.1	18.4	17.3	18.6	17.6	18.9	19.2
	3H	16.3	17.4	16.7	17.7	18.1	17.1	18.2	17.5	18.6	19.0
	4H	16.2	17.2	16.7	17.6	18.0	17.1	18.0	17.5	18.4	18.8
	6H	16.1	17.0	16.6	17.4	17.8	17.0	17.8	17.4	18.2	18.6
	8H	16.1	16.8	16.5	17.3	17.7	16.9	17.7	17.4	18.1	18.6
	12H	16.1	16.7	16.5	17.2	17.6	16.9	17.5	17.4	18.0	18.5
8H	4H	16.1	16.8	16.5	17.3	17.7	16.9	17.7	17.4	18.1	18.6
	6H	16.0	16.6	16.5	17.1	17.6	16.9	17.4	17.3	17.9	18.4
	8H	16.0	16.5	16.5	17.0	17.5	16.8	17.3	17.3	17.8	18.3
	12H	15.9	16.3	16.4	16.8	17.4	16.7	17.2	17.2	17.7	18.2
12H	4H	16.1	16.7	16.5	17.2	17.6	16.9	17.5	17.4	18.0	18.5
	6H	16.0	16.5	16.5	17.0	17.5	16.8	17.3	17.3	17.8	18.3
	8H	15.9	16.3	16.4	16.8	17.4	16.7	17.2	17.2	17.7	18.2

Rating by the formula method
For rooms with proportions (width-to-length ratios) that differ considerably from those listed in the tables (e.g. platforms), glare can also be rated using the UGR formula. This presupposes, however, that the position and viewing direction of the observer are known.

Current design software products offer direct UGR calculation and also an informative representation of UGR values for different observation angles.

Where direct rating is performed using the formula, even minor changes in the observer's position – e.g. 0.3 m – can result in variations of several tenths of a point. This often occurs where the intensity of light distributed by a lighting installation differs considerably across the beam (as in the case of specular louver luminaires, for example, or LED luminaires with lens optics). Where light distribution is uniform (e.g. luminaires with opal enclosures), however, observer positioning has little effect on UGR values. So a designer rating glare by the formula method needs to proceed with great care and attention to detail. Where light distribution is uneven, calculations should always be performed at a number of points to check the impact of variations in observer positioning.

Studies have shown that the formula method generally produces a glare prediction that corresponds closely to the subjective assessment of glare by test subjects. However, extensive experience of UGR limits is available only for the tabular method. For this reason and because of the impact of varying observer position, the only normative method recognised by DIN EN 12464 1 is the tabular method.

9.5 Appendix 5: Notes on maintenance factors

Maintenance factor is often abbreviated to MF. The abbreviations below are taken from CIE Publication 97.

Lamp lumen maintenance factor LLMF
As length of service increases, the lumen output of practically any lamp decreases as a result of ageing. How gradual and how pronounced that decrease is depends on the type and watt rating of the lamp in question and, where applicable, on the operating gear used. The ratio of luminous flux after a specific number of burning hours to the luminous flux when the lamp was new is indicated by the lamp lumen maintenance factor (LLMF).

LLMF values can be obtained from manufacturers or found in standard average value curves and lighting publications such as CIE Publication 97.

Lamp survival factor LSF
Each lamp in a lighting system has an individual life which is longer or shorter than the average service life. Average service life is the number of hours for which an observed group of lamps operate before half of the lamps fail. The probability that a relative set will still be operative after a specified number of burning hours is expressed by the lamp survival factor (LSF)

As with the lamp lumen maintenance factor, the magnitude and time-frame of the lamp survival factor depend on the type and watt rating of the lamp in question. In the case of discharge lamps, the LSF also depends on the operating gear used and the frequency of operation of the system.

In the case of fluorescent lamps, average service life is normally calculated on the basis of a switching rhythm of 2¾ h on / ¼ h off. With discharge lamps, the rhythm is 11.5 h on / 0.5 h off. LSF values are obtained from the same sources as LLMF values.

Luminaire maintenance factor LMF
Generally speaking, dirt deposited on lamps and luminaires causes a greater reduction of luminous flux than any other factor. The degree of light loss depends on the nature and particle size of the airborne pollutants, on the design of the luminaires and on the lamps used in them

CIE Publication 97 proposes a six-stage schematic type-coding common luminaires. Here, depending on luminaire type and accumulation of dust/dirt, luminaire maintenance factors (LMF) can be determined as a function of the time luminaires have spent in the lighting system since the last cleaning operation.

Room maintenance factor RMF
Dust deposits on ceiling, walls, floor and furnishings generally cause a reduction of indirect illuminance due to inter-reflection. The room maintenance factor takes account of the impact of these environmental conditions.

The room maintenance factor (RMF) can be defined as the ratio of utilance at a particular time to the utilance when the room surfaces were last cleaned.

Like utilance, the room maintenance factor basically depends on the size of the room, the reflectance of the room surfaces and the luminous flux distribution of the lighting system. In addition, the room maintenance factor depends on the type and amount of dirt in the air, which has a direct impact on the reduction of room surface reflectance. For simplified assumptions, standard RMF values can be found in CIE Publication 97.

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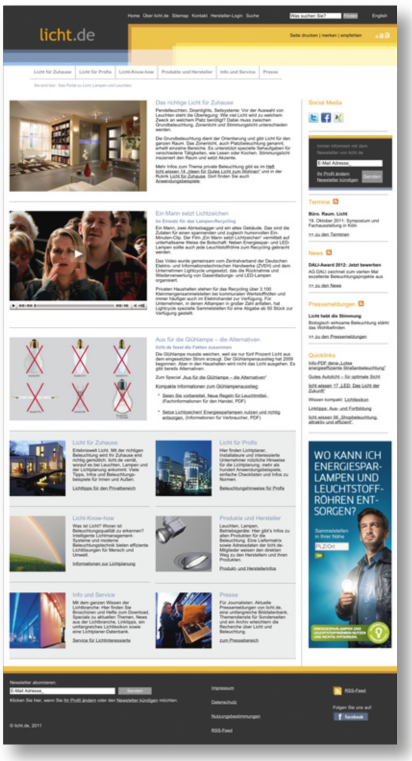

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